

BotanicAR: a cooperative experience in Augmented Reality

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ABSTRACT

We present BotanicAR, a collaborative Augmented Reality (AR) game leveraging the capabilities of Meta Quest 3. With Meta Quest 3, BotanicAR integrates both augmented and virtual reality in a singular device, ensuring a minimum synchronization of 72 frames per second (FPS). BotanicAR offers a multiplayer experience for 2 to 4 users who collaboratively undertake a task: fight against a virtual plant within their real environment. The game encourages natural interaction without the need for controllers, utilizing hand gestures for enhanced immersion. Real-virtual object occlusion further integrates game elements seamlessly into reality. Beyond its playful nature, BotanicAR demonstrates potential applications in collaborative training, emergency simulations, and shared augmented reality workspaces. This innovative application fills a gap in local cooperative experiences, pushing the capabilities of the Meta Quest 3 device to the limit.

Index Terms: Human-centered computing—Collaborative and social computing

1 INTRODUCTION

In recent years, modern technologies have become integral parts of our daily lives. One of the most significant recent developments is the spread of Augmented Reality (AR) across various sectors. Unlike Virtual Reality, AR does not replace physical reality but rather adds computer content to real environments. In fact, Milgram and Kishino conceive AR as the set of situations in which a real context is "augmented" by virtual elements, outlining a continuum between the physical environment and a purely virtual one with variable values of virtuality and reality [4]. AR is finding applications in various fields: education [5], training [3], culture [1], etc... Even in the increasingly dynamic and innovative video game landscape, AR is emerging as a revolutionary technology capable of radically transforming the gaming experience.

The authors in [2] created an augmented reality simulator for firefighting, to provide a realistic experience in a safe virtual envi-

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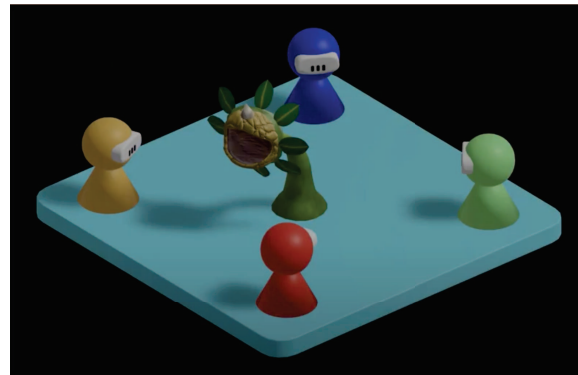


Figure 1: The plant materializes between the players

ronment. The authors use a mixed system with Google Pixel 6 and Microsoft HoloLens, they also focus a lot on the synchronization of avatars in the shared space guaranteeing at least 60 frames per second (FPS).

Thanks to the use of Meta Quest 3 we can integrate augmented reality and virtual reality in a single device, synchronization of at least 72 FPS is also ensured thanks to the use of Photon Fusion and the support of the OpenXR libraries.

In this demo paper, we present a collaborative augmented game. A number of users from 2 to 4 must complete a common task by collaborating with each other, specifically the task consists of fighting against a toxic plant, Figure 1. The environment that users will be placed in corresponds to their real environment, with the addition of a virtual plant that needs to be defeated.

The system can be used without the use of controllers so the immersion is amplified by the use of the hand. Furthermore, the occlusion of real objects with virtual ones allows users to fully integrate the elements of the game with those of reality. This demo shows a playful activity, but can also be adapted to create collaborative training, simulate emergency situations, or create shared workrooms in augmented reality.

Currently, there is no application with this level of local cooperation, also because the device used is very recent.



Figure 2: The game

2 SYSTEM DESIGN

The system is based on local multiplayer in a shared space. The features are based on the *Meta Quest 3* devices. The system is based on a Client-Server architecture and it was developed by using *Unity*, integrating different libraries: *Fusion*, for multi-user functionality, allowing multiple users to network to share the same virtual environment; *Photon-Voice*, which creates a voice communication channel allowing a shared voice chat system among all users. We used the *Meta Quest 3 Depth API*¹ to generate virtual objects in a way that makes them hidden behind real-world objects and surfaces, creating the illusion that they are seamlessly integrated into the actual environment. Thanks to this API, we were able to integrate an augmented reality occlusion system. Without these occlusions, the virtual content would be visibly distinct, appearing as a separate layer overlaid on the real world, disrupting the immersive experience of mixed reality (MR). *Spatial anchors*² act as fixed points of reference, enabling precise content positioning. These transformations represent stable physical locations in the real-world environment. Utilizing spatial anchors improves user experience by enabling activities such as attaching virtual signs to existing furniture or setting starting points on real windows for virtual characters to move through. Synchronization of at least 72 FPS is also ensured thanks to the use of Photon Fusion and the support of the *OpenXR* libraries. The *Oculus Interaction SDK* was used to enable interaction with the scenario and the various objects in it, but also to enable hand tracking without the use of controllers, through the use of HeadSet cameras. In addition, interactions with objects have been customized by taking advantage of code components developed in C#.

3 THE GAME

The game, shown in Figure 2, in which 2 to 4 players can participate, is organized into 3 levels and it consists of a plant, which is the central and vulnerable objective of the game. It is positioned in the center of the room. The players' main objective is to attack and defeat the plant, cooperatively. Around the plant, roots randomly emerge from the ground, creating moving obstacles for the players. If a player hits a root, he loses the object in his possession. Roots can have the following characteristics:

- some roots will have a bomb;
- other roots will contain a bottle of weed killer;
- one root will have a fire;
- *vulnerable root*: a specific root among those scattered around will be designated as a vulnerable root. This is the key target for players to plant the bomb;

¹<https://developer.oculus.com/documentation/unity/unity-depthapi/>

²<https://developer.oculus.com/documentation/unity/unity-spatial-anchors-overview/>

- *bomb*: when triggered and placed on the vulnerable root, the bomb will have a 3-second countdown before detonating, making the plant vulnerable to attack;
- *fire*: a fire will be available among the roots. The player who possesses it can light the bomb when it is placed on the vulnerable root.

During all levels, except the time when it becomes vulnerable, the plant will continuously attack, firing projectiles at players. When the plant is vulnerable, players can use herbicide to attack it and reduce its health.

Once the plant's health is reduced, players advance to the second level. Once the second level is completed, they access the third level, each with an increasingly challenging configuration of roots and objects.

4 CONCLUSION

In this demo, we present BotanicAR, a collaborative Augmented Reality (AR) game leveraging the capabilities of Meta Quest 3. BotanicAR offers a multiplayer experience for 2 to 4 users who collaboratively fight against a virtual plant within their real environment. Meta Quest 3 allows the integration of augmented reality and virtual reality in a single device, thanks to Photon Fusion and the support of the OpenXR libraries. BotanicAR encourages natural interaction with a fully hand-tracking experience. Also, real-virtual object occlusion further integrates game elements seamlessly into reality. The players can enjoy three levels, and beyond its playful nature, BotanicAR demonstrates potential applications in collaborative activities.

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